

Fundamental MATLAB Programming for AI

Today's goals

Module 1The Fundamentals of MATLAB

Module 2Produce Effective MATLAB Programs for Science and Engineering Problems

Module 2 Introduction

- Introduction to Module 2's StudyVideo • 50 sec

Module 2-1 Custom MATLAB Functions

- Custom MATLAB Functions: Creating Custom MATLAB FunctionsVideo • 9 min
- Custom MATLAB Functions: Anonymous FunctionsVideo • 7 min
- Materials about Custom and Anonymous FunctionsReading • 10 min

Module 2-2 MATLAB Conditionals and Loops

- MATLAB Conditionals and Loops: Conditionals and Program Control in MATLABVideo • 7 min
- MATLAB Conditionals and Loops: Loop in MATLAB ProgrammingVideo • 7 min
- Materials about Conditionals and LoopsReading • 10 min

Module 2-3 Case Study: Kepler's Third Law

- Case Study: Kepler's Third LawVideo • 8 min
- Materials about Kepler's Third LawReading • 10 min

Module 2-4 Case Study: The Newton Raphson Method

- Case Study: The Newton Raphson MethodVideo • 6 min
- Materials about Newton Raphson MethodReading • 10 min
- Designing Reliable MATLAB Programs for Science and EngineeringDialogue • 5 min

Module 2 Quiz

- Module 2 QuizPractice Assignment • Grade: 100%

Module 2 Practice/Warm-up Exercise for Graded Assignments

- Module 2 Exercise 1Ungraded App Item • 30 min
- Module 2 Exercise 2Ungraded App Item • 30 min
- Exercise SolutionsReading • 10 min

Module 2 Assignment

- Module 2 Assignment 1Graded App Item • 30 min
- Module 2 Assignment 2Graded App Item • 3 min
- Module 2 RecapReading • 5 min

Module 2 Recap

0:02 / 0:41

2x

In this module, you learned how to:

- Break large tasks into **custom functions**
- Control program behavior using **if / elseif / else**
- Perform repeated operations with **for** and **while** loops
- Combine logic, iteration, and modular design to build structured engineering programs
- Apply these skills in case studies such as **Kepler’s Law** and **Newton–Raphson**, where MATLAB was used to solve real scientific problems

At this stage, you should be able to:

- Design reusable functions instead of writing long scripts
- Implement decision-making and stopping criteria
- Build iterative numerical methods
- Translate mathematical models into reliable MATLAB code

This is an important transition point.

Go to next item

✓ Completed

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